

MARKING SCHEME

SOFTWARE CAT3

MODULE 3 TERM 3

Question 1: Define the term software maintenance. (2 Marks)

Answer:

Software maintenance is the process of modifying, updating, repairing, and improving software after installation to ensure it continues to operate effectively and efficiently.

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- Correct definition – 2 Marks

Question 2: State two importance of software maintenance. (2 Marks)

Any Two

- Corrects software errors and bugs.
- Improves software performance.
- Enhances system security.
- Improves reliability.
- Extends software lifespan.
- Ensures compatibility with new hardware and software.

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- 1 Mark each $\times 2 = 2$ Marks

Question 3: List two activities that may be included in a software maintenance schedule. (2 Marks)

Any Two

- Installing software updates.
- Applying security patches.
- Running antivirus scans.
- Backing up data.
- Disk cleanup.
- Monitoring software performance.

- Removing obsolete software.

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- 1 Mark each $\times 2 = 2$ Marks

Question 4: Differentiate between: (4 Marks)

a) Corrective Maintenance

Maintenance carried out to identify and fix software faults, bugs, or errors after they occur.

b) Preventive Maintenance

Maintenance performed to prevent future software failures and improve software reliability before problems occur.

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- Corrective maintenance explanation – 2 Marks
- Preventive maintenance explanation – 2 Marks

Question 5: Match the following software maintenance types with their descriptions. (4 Marks)

Maintenance Type	Correct Description
a) Adaptive Maintenance	iv. Modifying software to operate in a new environment
b) Perfective Maintenance	i. Improving software performance and usability
c) Preventive Maintenance	iii. Preventing future software failures
d) Corrective Maintenance	ii. Correcting software faults and errors

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- 1 Mark for each correct match $\times 4 = 4$ Marks

Question 6: State any three types of software updates. (3 Marks)

Any Three

- Security updates.
- Feature updates.
- Bug-fix updates.
- Driver updates.
- Firmware updates.
- Service packs.

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- 1 Mark each $\times 3 = 3$ Marks

Question 7: State three Personal Protective Equipment (PPE) used when handling computer equipment. (3 Marks)

Any Three

- Antistatic wrist strap.
- Safety goggles.
- Protective gloves.
- Dust mask.
- Antistatic mat.

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- 1 Mark each $\times 3 = 3$ Marks

Question 8: Explain four importance of software upgrades. (4 Marks)

Any Four

- Improve software performance.
- Enhance security.
- Fix bugs and software errors.
- Add new features and functions.
- Improve compatibility with newer systems.
- Increase software stability and reliability.

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- 1 Mark each explanation $\times 4 = 4$ Marks

Question 9: Describe three types of software upgrades. (6 Marks)

i) Security Upgrade

Updates designed to protect software against security threats and vulnerabilities.

ii) Feature Upgrade

Updates that introduce new features and functionalities.

iii) Performance Upgrade

Updates intended to improve software speed, efficiency, and stability.

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- Identification of type – 1 Mark
- Correct description – 1 Mark

(2 Marks $\times 3 = 6$ Marks)

Question 10: Define software functionality monitoring. (2 Marks)

Answer:

Software functionality monitoring is the continuous observation and assessment of software performance and operation to ensure it functions as intended.

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- Correct definition – 2 Marks

Question 11: Explain the following types of operating system event logs. (8 Marks)

a) Error Event Logs

Record serious problems that prevent software, applications, or system components from functioning properly. (2 Marks)

b) Warning Event Logs

Record events indicating potential issues that may cause future problems if ignored. (2 Marks)

c) Information Event Logs

Record successful execution of normal system operations and activities. (2 Marks)

d) Success Audit Event Logs

Record successful security-related activities such as user logins and authorized resource access. (2 Marks)

Total: 8 Marks

Question 12: Prepare a simple monthly software maintenance schedule for a computer laboratory showing four maintenance activities. (4 Marks)

Sample Answer

Week	Maintenance Activity
Week 1	Update operating system
Week 2	Run antivirus scan
Week 3	Backup laboratory data
Week 4	Disk cleanup and software performance monitoring

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- Any relevant maintenance activity correctly scheduled – 1 Mark each $\times 4 = 4$ Marks

Question 13: Explain three importance of software updates. (6 Marks)

Any Three

- Improve system security by patching vulnerabilities.
- Correct software bugs and errors.
- Improve software performance and stability.
- Enhance compatibility with hardware and software.

- Provide new features and functionality.

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- Correct explanation – 2 Marks each $\times 3 = 6$ Marks

Question 14: Explain three safety measures observed during software maintenance activities. (6 Marks)

Any Three

- Back up important data before maintenance to prevent data loss.
- Use genuine and trusted software tools and updates.
- Ensure antivirus protection is active before installing updates.
- Follow organizational maintenance procedures.
- Verify software sources before downloading updates.
- Restrict unauthorized access during maintenance.

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- Correct explanation – 2 Marks each $\times 3 = 6$ Marks

Question 15: Explain the importance of proper use of tools and equipment when carrying out software maintenance. (4 Marks)

Any Four

- Prevents damage to computer systems.
- Improves maintenance accuracy.
- Increases work efficiency and productivity.
- Enhances safety of personnel and equipment.
- Reduces maintenance costs.
- Ensures quality and reliable maintenance outcomes.

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- 1 Mark each $\times 4 = 4$ Marks